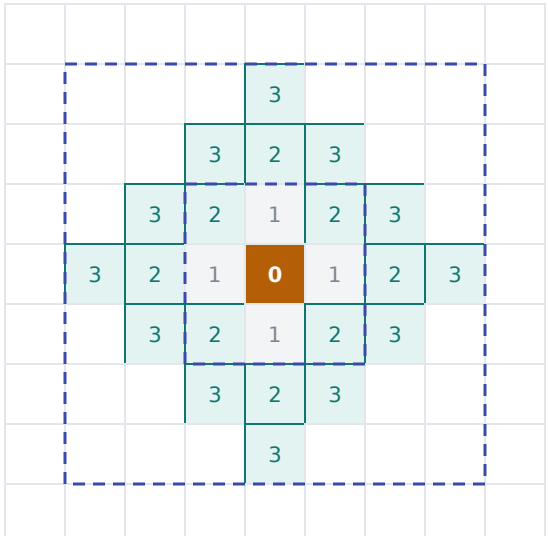
The ring of a one-pixel object: 4-neighbour dilation, steps 2-3



the object (distance 0)

distance 1: 4 pixels, skipped (too close)

distance 2: 8 pixels; distance 3: 12 pixels

ring = 8 + 12 = 20 pixels

a square ring (3 x 3 dilation) would be 7 x 7 minus 3 x 3

= 49 - 9 = 40 pixels - not what the code builds

ring = binary_dilation(obj, iterations=3) & ~binary_dilation(obj, iterations=1) & valid

scipy's default structure is the 4-neighbour cross: steps = rows apart + columns apart.

Real and implanted objects use the same ring - the tier edges depend on it.

tests/test_utils/test_implant.py, test_contrast_z_by_hand: ring.sum() == 20.